context for new conversations. On the other hand, human demonstrators actively provide in-context demonstrations to enhance the reasoning capabilities of models. However, these demonstrations, whether crafted by humans or sourced from datasets, can also contain noisy rationales.

Irrelevant thoughts - model perspective: LLMs tend to explain the concept of terms during reasoning. For instance, if you ask GPT-4 to debug an error related to the 'concurrent' package, it tends to start by explaining what the 'concurrent' package is rather than immediately addressing the debug request. Such explanations, while informative, may introduce irrelevant noise into the conversation. Here are some key reasons for their emergence.

1. **Cognitive bias of models.** When addressing complex queries, LLMs tend to include explanations for terms or concepts mentioned in the prompt that are unnecessary for solving the specific problem presented, as a kind of irrelevant thought. This behavior exhibits a form of cognitive bias where the model, unable to assess the inquirer's level of understanding, leads to explicating background information. This is comparable to a lecturer explaining the basics of a subject to a class without first assessing their students' existing knowledge, potentially leading to irrelevant elaborations.
2. **Lack of precise contextual understanding.** Despite LLMs' proficiency in processing language and recognizing patterns, they cannot always fully grasp the precise context or specific requirements of a problem. This shortfall can result in the production of thoughts that, although seemingly related, do not directly contribute to resolving the question at hand. Their responses might align more with the broader theme of the discussion rather than the specific, critical details needed for a precise solution.
3. **User query ambiguity of the dialogue mechanism.** Ambiguities in user queries can stem from the use of vague or multifaceted language, prompting the model to generate a wide array of responses. This situation is similar to a search engine returning a variety of results for a query that lacks specificity. The model, attempting to cover all potential meanings of the query, may produce responses that contain thoughts unrelated to the user's actual intent. For instance, if someone asks, "What is force calculation?", the model might provide information on both Newtonian mechanics and quantum mechanics. These responses, encompassing a broad range of topics, could influence the answers to subsequent physics questions, leading to a continuation of the ambiguity and further complicating the conversation.
4. **Progressive disclosure bias of the dialogue mechanism.** When engaging with LLMs, individuals often initiate the dialogue by describing simpler concepts and progressively work their way toward articulating the complex question at hand because of human limitations in language or comprehension abilities, which prevent a person from presenting the entire complexity of an issue in one go. This step-by-step approach, while natural for humans who struggle to directly convey intricate problems, can introduce extraneous content that contributes to noise within the model's contextual reasoning. As the conversation builds, the LLM will factor in these initial, possibly tangential, explanations into its understanding of the context, potentially leading to a dilution of the focus necessary for solving the specific issue. This phenomenon reflects a human cognitive strategy in communication that may not be optimally aligned with the operational mechanisms of LLMs for efficient problem-solving.

Irrelevant thoughts - human perspective: When a human is actively demonstrating CoT reasoning, the introduction of irrelevant thoughts could be due to a variety of reasons.

1. **Cognitive overload of humans.** Humans may introduce irrelevant information when they are trying to process too much information at once, which can lead to a loss of focus and the inclusion of tangential thoughts. For instance, a programmer is struggling with a bug in a complex piece of software and asks an LLM for help. To provide context, the programmer starts explaining the issue with a CoT rationale approach, intending to walk the LLM through their thought process. However, due to the complexity of the code and the stress of finding the bug, the programmer gets sidetracked. They include unnecessary details about the different error messages encountered in the past, unrelated functions in the code, and general thoughts on software development.
2. **Associative thinking of humans.** Humans naturally think in an associative manner, where one thought may lead to another that is only loosely related to the task at hand. This can result in straying from the main point during a CoT explanation. For example, while a programmer outlines the steps to diagnose a software issue for an LLM, they might recall a

similar problem encountered in a different project. This memory could lead them to mention troubleshooting strategies, tools, or anecdotes from that experience, which, although related to the broader theme of problem-solving, do not directly contribute to the current issue.

3. **Irrelevant content in datasets.** In the future, it is likely that companies or professional organizations will increasingly utilize databases to assemble CoT prompts. However, these databases, whether privately maintained or publicly accessible, can contain irrelevant reasoning processes. This is especially true for databases sourced from crowdsourcing platforms or open forums, where the information is contributed by a diverse set of individuals with varying levels of expertise and focus. When these datasets are used to provide in-context information for CoT reasoning, the noise can originate from the inclusion of off-topic discussions, personal opinions, or overly verbose explanations that do not directly address the problem at hand. Such noise can be inadvertently introduced into the CoT process when humans provide explanations that contain unnecessary or tangential information.

Similarly, we analyze the two sources of inaccurate thoughts as follows.

Inaccurate thought - model perspective: For models, LLMs may produce erroneous thoughts during the reasoning process, especially when dealing with complex problems. For example, when tackling a base-9 math problem in a zero-shot setting, GPT-3.5 may generate some inaccurate reasoning steps. The former dialogue will become inaccurate and noisy in the context of subsequent dialogues. Here are some key reasons for their emergence.

1. **Outdated or incomplete training data of the model.** Language models are built upon datasets that may not be current or fully comprehensive. When faced with problems that require up-to-date knowledge or complete understanding, which are absent in their training data, models may rely on outdated or incomplete information, resulting in inaccurate outputs. For example, in the field of medicine, if new research suggests a change in treatment protocol after the model's last update, it wouldn't be able to advise on the new information.
2. **Adaptation to novel reasoning contexts of the model.** New challenges may require models to reason within contexts that slightly or significantly differ from their training data. For instance, a model extensively trained on base-10 arithmetic might struggle with a base-9 math problem because it requires a shift in the underlying numerical framework. This kind of scenario demands on-the-fly adaptation to a novel reasoning context, which can lead to generating thoughts that do not accurately apply the learned principles from the base-10 system to the newly introduced base-9 system.
3. **Misinterpretation of complex subjects of dialogue mechanism.** Users often fail to clearly articulate their complete requirements at the outset of an inquiry, leading to LLMs generating misunderstandings and inaccurate thoughts that do not align with user expectations. The process of correcting these thoughts is inherently a reasoning process laden with noisy contexts. As users provide feedback to refine the model's output, the iterative nature of this interaction can introduce additional inaccuracies as the model attempts to reconcile the new information with the previously misunderstood context.

Inaccurate thought - human perspective: Inaccurate thoughts in CoT can stem from the information provided by humans, whether it is self-made on the spot or sourced from a database for in-context learning by LLMs. These CoT demos can include inaccurate noise due to various factors.

1. **Personal knowledge limitations of human.** Individuals may possess incomplete or outdated knowledge on a given subject, leading to the provision of incorrect information when creating a CoT. For instance, a person without expertise in mathematics might attempt to construct a CoT for a complex math problem and inadvertently introduce incorrect steps or conclusions. Their understanding may be based on heuristics or educational background that hasn't been updated to reflect more recent methodologies or discoveries in the field.
2. **Cognitive biases of human.** Human reasoning can be influenced by a range of cognitive biases, such as confirmation bias, where an individual tends to search for, interpret, and remember information in a way that confirms their preconceptions, neglecting contrary information. Or the oversimplification of complex issues might lead to inaccurate reasoning steps within a CoT. These biases can skew the logic flow and result in conclusions that do not hold up under scrutiny or are based on flawed premises.

3. **Data quality issues of database.** The databases that humans rely on for creating CoTs might contain errors or biases introduced during data collection and processing. If this flawed data is used for in-context learning by LLMs, it can impart incorrect patterns of thought or factual inaccuracies. For example, a dataset with biased sampling methods might lead to generalizations that do not accurately represent the broader population or situation.
4. **Contextual misplacement of databases.** Information from databases may be stripped of its original context, leading to misinterpretation when reused. When humans include such decontextualized information in a CoT, they might not properly align it with the new context, introducing misunderstandings or inaccuracies. This is particularly problematic in nuanced fields where context heavily influences the meaning and applicability of information, such as legal precedents or cultural studies.

Given the convenience and adaptability of CoT reasoning, broader adoption in LLM applications is expected in the future. This structured approach enables LLMs to break down complex problems and explain their reasoning in a way that resembles how humans think, proving essential for sophisticated problem-solving. Nonetheless, we are bound to face the noisy reasoning challenges, stemming from both model-generated and human-contributed contexts, as mentioned above.

To address these challenges, we must focus on continuously improving training methods, keeping models updated with the latest information, enhancing their ability to parse context and ambiguity, and refining algorithms to diminish biases and logical inaccuracies.

Differences between inaccurate and irrelevant thoughts. Given the context and question, a thought is either relevant or irrelevant. Within the relevant thoughts, those accurate ones are desirable, and those inaccurate, termed as "inaccurate thoughts" in this work, are proven to be harmful to LLMs. Hence, the extreme case of "inaccurate and irrelevant thought" is not covered in this work. For clarity and simplicity, we investigate these two kinds of noise separately. Specifically, in definitions,

- Irrelevant thoughts refer to incorporating extraneous details that are unhelpful for solving the question. Redundant information may be introduced by the LLM's diverse response generation or by humans when clarifying concepts in problem-solving examples;
- Inaccurate thoughts refer to factual errors in rationales that are common in mathematical calculation or transcription. The emergence of noise can be due to algorithmic limitations, errors in training data, misinterpretations of context or instructions, and logical fallacies.

Note that the "accuracy" and "relevance" of thoughts are related to the context of the given question. Basically, the question provides the context, and an LLM generates thought conditional on the context. Taking the examples of the Base-9 dataset in Tab. 1,

- Irrelevant thought is "There are five oceans on Earth: the Atlantic, Pacific, Indian, Arctic, and Southern.", which is accurate but not relevant and not useful;
- Inaccurate thought here is "5 + 9 = 14", which is relevant to the question and previous thought but is inaccurate for the base-9 calculation.

In the NoRa dataset, only the relevant thought with factual error will be classified as an "inaccurate thought". There is no irrelevant and inaccurate thought in NoRa, which should be rare in practice.

We focus on the two major types of noise in this work. And empirically, inaccurate thoughts bring severe degradations. Compared with clean rationales, a 1.4%-19.8% decrease with irrelevant noise and a more drastic 2.2%-40.4% decrease with inaccurate noise.

C.3 Real-world Examples

The emphasis on noisy rationale is due to its practical challenges, with examples drawn from diverse sources such as crowdsourced platforms, dialogue systems, and AI-generated data. Here, the Noisy-R mainly originates from (1) the inherent imperfections, inconsistencies, and inaccuracy of humans' cognitive processes and (2) the diversity, unpredictability, and hallucination of the LLMs' generative mechanisms, as discussed in more detail in Appendix C.2.

Briefly, irrelevant and inaccurate thoughts can be generated by both the model and humans. From model perspective, the generated rationales by the model can be noisy. From human perspective, the annotated rationales by humans can be noisy.